Evaluation 1: Corpus Confirmation and Project Plan

Mini Project 14: Pharmacovigilance Adverse Event Intake and Triage Service · UE24AM342AA5 ML System Design & AgentOps
Team: PES1UG24AM287 · PES1UG24AM285 · PES1UG24AM265 · PES1UG24AM258 · **Date:** 30 October 2026 · **Repository:** https://github.com/srivamshichikkireddi/PharmacovigilanceMLSD

> *Everything in sections 2-4 is generated from the real corpus by* `scripts/eval1_evidence.py` *and saved in* `evaluation1/evidence/`*. Re-run it to verify any claim:* `python -m pv.corpus.build && python scripts/eval1_evidence.py`.

1. Summary

We confirm a three-document corpus (all public, all with a text layer, 1,217 chunks), map every passage the section 4.1 case depends on to a document and page, state the values we will test with, and show the baseline: a single retrieval pass using the reporter's own words retrieves **0 of the 6** supporting pages for the 4.1 case, which is exactly the failure the brief predicts. One point needs the instructor's confirmation at this evaluation: the reference route for a serious but *expected* reaction (section 3.4).

2. Finalized corpus

File stem	Document	Source	PDF pages	Chunks	SHA-256	Role
`cdsco_rules`	New Drugs and Clinical Trials Rules, 2019 (CDSCO), incl. Fifth Schedule	cdsco.gov.in	264	951	`b25fdd5476a2`	Law: safety update reports and the serious-unexpected exception
`ipc_mah_v2`	Pharmacovigilance Guidance Document for Marketing Authorisation Holders (IPC, Sept 2024)	ipc.gov.in	86	171	`9142de7460f7`	Operating guidance: valid ICSR, seriousness criteria, timelines
`tegretol_us_pi`	Tegretol (carbamazepine) prescribing information (Novartis, US)	novartis.com	25	95	`7e788668f892`	Label: which reactions are listed (expectedness)

- **Total:** 1,217 chunks; corpus version `17027899456b`; index built in 0.98 s (1236 chunks/s) by an idempotent CLI pipeline (`python -m pv.corpus.build`; a re-run skips unchanged documents). No OCR needed: all three have a text layer.
- **Drug: carbamazepine (Tegretol).** Its label carries a boxed warning for serious dermatologic reactions, so the 4.1 reaction is a *listed* one under medical names. Adapting the case to another drug needs only a new insert PDF and one entry in `corpus/drugs.json`.
- **Not indexed: WHO-UMC causality criteria.** The system assesses seriousness, expectedness and route; it makes no causality claim.
- **Limitation we state openly:** the insert is the US label, not an Indian-market one.
- **Extraction decision (measured):** `pypdf` inserted stray spaces inside words on the Gazette PDF and returned about four times more characters per page than the real text. We use poppler's `pdftotext` (extractor recorded in the manifest), with `pypdf` as fallback.
- **Synthetic data only.** No real patient or case data enters the system.

3. Section 4.1 mapping

> "*The patient's skin started peeling badly a week after starting the drug, and she was admitted to hospital.*"

3.1 What the report says, and what the documents call it

Reporter's words	Entity	Document language it must be matched to
"skin started peeling badly"	reaction	exfoliative dermatitis; toxic epidermal necrolysis; Stevens-Johnson syndrome
"admitted to hospital"	outcome / seriousness	hospitalisation criterion (IPC 2.5.3.6(iii))
"a week after starting"	temporal sequence	onset latency 7 days; checked against the two dates
"the drug"	suspect drug	carbamazepine (in corpus: yes)

3.2 Passages the case depends on (each located by anchor text and verified in the corpus)

#	Role	Provision	Document, PDF page	Excerpt (short; full text on the page)	Verified
1	Label: boxed warning names the reaction	Stevens-Johnson syndrome (boxed warning)	tegretol_us_pi p.1	"STEVENS-JOHNSON SYNDROME (SJS), HAVE BEEN REPORTED DURING TREATMENT WITH TEGRETOL. THESE REACTIONS ARE ESTIMATED"	yes
2	Label: warnings section	toxic epidermal necrolysis	tegretol_us_pi p.1, 4	"TOXIC EPIDERMAL NECROLYSIS (TEN) AND STEVENS-JOHNSON SYNDROME (SJS), HAVE BEEN REPORTED DURING TREATMENT WITH"	yes
3	Label: adverse reactions, skin	exfoliative dermatitis	tegretol_us_pi p.13	"exfoliative dermatitis, erythema multiforme and nodosum, purpura, aggravation of disseminated lupus erythematosus, alopecia, diaphoresis,"	yes
4	Seriousness criterion: hospitalisation	IPC 2.5.3.6(iii)	ipc_mah_v2 p.31	"Hospitalization /prolongation of existing hospitalization: If Adverse Events 20 caused hospitalization or increased the"	yes
5	Timeline: serious events	IPC 2.7.1	ipc_mah_v2 p.33	"All Serious Adverse Events must be reported by MAH within 15 calendar days of"	yes
6	Timeline: non-serious events (contrast)	IPC 2.7.2	ipc_mah_v2 p.33	"All Non-Serious Adverse Events must be reported by MAH within 90 calendar days of"	yes
7	Provision the request never mentions: serious UNEXPECTED reactions	NDCT Fifth Schedule	cdsco_rules p.229	"all cases involving serious unexpected adverse reactions must be reported to the licencing authority"	yes
8	PSUR cycle the exception modifies	NDCT Fifth Schedule	cdsco_rules p.229	"Periodic safety update reports due for a period must be submitted within thirty calendar"	yes

3.3 Test values


```
{
  "patient_age": 35,
  "patient_sex": "female",
  "suspect_drug": "Tegretol (carbamazepine)",
  "start_date": "2026-10-01",
  "reaction_date": "2026-10-08",
  "received_date": "2026-10-26",
  "hospitalization": true,
  "death": false,
  "life_threatening": false,
  "disability": false,
  "congenital_anomaly": false,
  "outcome": "unknown",
  "reporter_type": "doctor",
  "reaction_description": "The patient's skin started peeling badly one week after starting the drug, and she was
      admitted to hospital."
}
```


Onset latency is 7 days (consistent with "a week"); the clock starts at `received_date` (26 Oct 2026). A further 25 scenarios cover other seriousness criteria, listed and unlisted reactions, non-serious cases, every escalation path and boundary rejections (`eval/scenarios.json`).

3.4 Reference answer: for instructor confirmation

Established with confidence: seriousness = **serious** (hospitalisation); expectedness = **expected** (listed on the label, pages 1, 4, 13).

The open point. The route for *serious* + *expected* depends on how two provisions interact:

- **IPC guidance 2.7.1 (p.33):** all serious adverse events are reported within 15 calendar days; expectedness is not mentioned.
- **NDCT Fifth Schedule (p.229):** safety update reports are periodic; however, serious *unexpected* reactions go to the Licensing Authority within fifteen days.

Option	Route for serious + expected	Rests on
A	Expedited, 15 days	IPC 2.7.1 read as applying to all serious events
B	Periodic safety report	NDCT p.229 read as reserving expedited reporting for serious *unexpected* reactions

Current implementation: Option A, with the NDCT/IPC difference flagged in the output and QPPV confirmation required; changing to B is a single branch in `decide()` and the evaluation expectations regenerate from one script.

Instructor confirmation: ☐ Option A (expedited) ☐ Option B (periodic) ☐ Other: _____ Confirmed by: _____ Date: _____

Decision table once confirmed:

Seriousness	Expectedness	Route	Basis
serious	unexpected	expedited, 15 days	IPC 2.7.1 and NDCT p.229
serious	expected	**per confirmation above**	IPC 2.7.1 / NDCT p.229
non-serious	either	periodic (90 days; PSUR cycle)	IPC 2.7.2 p.33; NDCT p.229
no valid ICSR (no adverse event described)	n/a	non-reportable, with reason	IPC 2.5 p.29

4. Baseline setup

Definition. The brief's "straightforward implementation": embed or search once with the request, take the top passages, answer from them. Implemented as a switch in the same codebase (`AGENTIC_RETRIEVAL=false`), with the same corpus, verifier and scenarios, so the comparison isolates retrieval strategy.

4.1 Baseline on the 4.1 case

Query: the report sentence plus the drug name. Top 8 returned: tegretol_us_pi p.20; tegretol_us_pi p.6; tegretol_us_pi p.6; tegretol_us_pi p.20; tegretol_us_pi p.22; tegretol_us_pi p.22; cdsco_rules p.262; tegretol_us_pi p.11.

- Needed: cdsco_rules p.229, ipc_mah_v2 p.31, ipc_mah_v2 p.33, tegretol_us_pi p.1, tegretol_us_pi p.4, tegretol_us_pi p.13.
- **Retrieved: 0 of 6** (recall 0.0). The NDCT p.229 provision is **not retrieved**; no IPC page was returned at all.
- Outcome: the verifier finds the evidence insufficient, so the run **escalates** with no citations. The baseline is safe but cannot answer, which is the failure the design approach describes.

4.2 Baseline on the full evaluation set (26 scenarios: 22 full cases, 4 boundary rejections)

Metric	Baseline: single pass	Agentic (for reference)
Route accuracy	0.32	1.00
Answered rather than escalated	0.00	0.68
Context recall of gold passages	0.09	0.94
Context relevance	0.04	0.37
Hallucination rate	0.00	0.00

Metric definitions are in `docs/EVALUATION.md`. Caveat: the scenarios and system were written by the same team; the gold labels need a second reviewer, and the 4.1 reference route is pending section 3.4.

4.3 Working end-to-end path

Per the team's checkpoint C2, a clean clone with `docker compose up` served the 4.1 case through the web client as an expedited recommendation with the conflict flagged and QPPV confirmation required. Evidence: `demo-result.png` [attach]. Environment: Python 3.11 in Docker; poppler `pdftotext` 24.02; Redis 7; FastAPI; LangGraph.

4.4 Reproduce the baseline

```
python -m pv.corpus.build                                    # idempotent index
python scripts/eval1_evidence.py                             # sections 2-4.1 evidence
python -m pv.eval.run --validate-gold                        # every gold passage exists
AGENTIC_RETRIEVAL=false python -m pv.eval.run --label single_pass_baseline
```

5. Project plan

5.1 Priorities

Phase	Focus	Requirements	Status
1. Working path	Pipeline, FastAPI and queue, agents, web client, compose up	D2, D3, A4, A8, F1	Done (C0-C2)
2. Correctness	Validation, verification, escalation, 4.1	D1, A1-A3, F5, F7	Built; reference route awaiting confirmation (C3)
3. Trust	Eval set, prompt versions, injection defence	L2, L3, L4	Eval set and before/after done; prompt comparison needs a model key
4. Operations	Tracing, Locust, breaker, cache, runbook	D5-D8, A5, A7	Written; verification and measurement in progress
5. Stretch	Cost report, feedback, live scoring	L5, L6	Built; measurement pending

5.2 Checkpoints

Checkpoint	Goal	Status
C0-C2	Offline baseline; clean-clone Docker start; functional checks in containers	Done
C3	Instructor confirms the 4.1 reference route	**This evaluation**
C4	Prometheus, Grafana, Jaeger verified end to end	Next
C5	Connect a model (spending limit set); compare two prompt versions	After C3
C6-C7	Retrieval precision; review of gold labels	Planned
C8	SLO committed before any load test	Before C9
C9	Locust at 1 and 3 workers; failing concurrency; bottleneck	Planned
C10-C11	Resilience drills; cache and cost measurement	Planned
C12-C13	Final report and README test; demonstration rehearsal	Planned

5.3 Declared SLO (set before any measurement)

p95 end-to-end latency (submission to terminal result) of 30 s or less at 1.0 runs/s or more with 3 workers and no failed runs, with the model replaced by a stub. Justification: regulatory clocks are in days, so the target protects the triage session and burst isolation, while the no-loss queue protects the deadline.

5.4 Roles

Four roles, to be assigned among the team: platform, deployment and load testing · agents, retrieval and evaluation · documentation and report · testing, review and demonstration rehearsal. Assignment: [fill in]. Every member must be able to explain the whole system.

5.5 Risks

Risk	Mitigation
Reference route for serious + expected	Confirmation requested here; one-branch change
Free-tier model limits	Spending limit; model behind a configurable interface; rules-only mode needs no key
Optional components not yet verified (vectors, tracing, dashboard)	Each degrades safely; verified at C4-C6
US label rather than an Indian insert	Stated limitation; one PDF and one config entry to change